

Consumer Complaint Intelligence & Risk Analytics Platform

Full Project Report

62,516 Consumer Financial Complaints · 2017–2023

SQL · Python · Power BI · Excel

Prepared by: Seema Kumari

Data Analyst | kriseema87@gmail.com

linkedin.com/in/seema-kumari-375763308 · github.com/seema-kri

1. Project Overview

This report presents a comprehensive analysis of 62,516 consumer financial complaints filed between 2017 and 2023. Using an end-to-end analytics pipeline — from raw data cleaning in Excel and Python through structured SQL analysis to executive Power BI dashboards — the project identifies high-risk products, geographic hotspots, operational gaps, and strategic improvement opportunities for financial institutions.

Business Problem

Financial institutions receive thousands of consumer complaints monthly across products, states, and channels. Without structured analytics, customer pain points remain hidden, high-risk products go undetected, and response performance lacks a measurable baseline. This platform answers the central question:

■ Which products, issues, and regions pose the highest risk — and is the organization responding effectively?

Stakeholders

Team	Primary Interest
Customer Service	Complaint volume, customer issues
Operations	Response efficiency, resolution effectiveness
Risk Management	High-risk products, complaint trends
Executive Management	KPIs, business performance overview

2. Solution Architecture

Layer	Tool	Key Activities
Data Ingestion	Excel	Raw CSV intake, duplicate removal, null standardization
Feature Engineering	Excel + Python	complaint_month, complaint_year, complaint_quarter, resolution_day
Data Wrangling	Python / Pandas	Dtype normalization, date parsing, outlier flagging
Database Layer	PostgreSQL 15	DDL schema, data load, structured query layer
Analysis	SQL (5 modules)	Validation → Business Analysis → KPIs → Window Functions
Visualization	Power BI	KPI scorecards, trend lines, risk ranking, channel donut, slicers
Documentation	Markdown	Project charter, KPI definitions, data dictionary, insights report

■ *SQL query suite spans 5 files covering DDL, data validation, business analysis, KPI computation, and advanced window functions (DENSE_RANK, SUM OVER, CTEs).*

3. Executive KPI Summary

TOTAL COMPLAINTS 62,516 2017–2023 cumulative	TIMELY RESPONSE 93.77% Benchmark: 90% ✓	LATE RESPONSE 3.84% Below 5% threshold	STATES COVERED 51 Full geographic coverage
PRODUCT CATEGORIES 9 Risk-ranked	ISSUE CATEGORIES 76 Classified	PEAK YEAR 2022 12,953 complaints	TOP PRODUCT SHARE 39.69% Checking & Savings

■ **Strong operational performance: 93.77% timely response rate exceeds the 90% industry benchmark. However, complaint volume is heavily concentrated in 2 product categories driving 65.6% of all complaints.**

4. Product Risk Analysis

4.1 Complaint Distribution by Product

Risk Rank	Product	Complaints	Share
#1 ■	Checking or Savings Account	24,814	39.69%
#2 ■	Credit Card or Prepaid Card	16,197	25.91%
#3 ■	Credit Reporting Services	7,710	12.33%
#4 ■	Mortgage	6,601	10.56%
#5 ■	Money Transfer Services	3,453	5.52%
#6 ■	Debt Collection	2,736	4.38%

Key Findings

- Nearly 40% of all complaints originate from Checking and Savings Accounts.
- The top 2 product categories account for 65.6% of total complaints.
- Deposit accounts and card products represent the largest operational risk areas.
- Complaint concentration is highly skewed — the bottom 4 products share only 32.79%.

■ **Business Recommendation:** *Prioritize customer experience improvements in account management and card services — these two products drive two-thirds of all customer dissatisfaction.*

5. Customer Issue Analysis

5.1 Checking & Savings Accounts

Issue	Complaints
Managing an Account	15,109
Closing an Account	2,953
Opening an Account	2,725
Unauthorized Charges	2,493

Over 60% of checking account complaints relate to account management friction. Account open/close/manage processes are the single largest customer experience challenge.

5.2 Credit Card & Prepaid Card

Issue	Complaints
Purchase Disputes	4,415
Getting a Credit Card	1,867
Product Terms & Conditions	1,633
Fees and Interest	1,422

Transaction disputes and onboarding issues dominate credit card complaints, signalling acquisition funnel friction and post-purchase billing confusion.

5.3 Credit Reporting Services

Issue	Complaints
Incorrect Information	4,145
Investigation Problems	1,641
Improper Use of Reports	1,517

5.4 Mortgage

Issue	Complaints
Trouble During Payment Process	2,827
Struggling to Pay Mortgage	1,904
Mortgage Application Process	1,017

5.5 Money Transfer Services — Fraud Alert

Issue	Complaints
Fraud or Scam	1,951

Issue	Complaints
Transaction Problems	571
Money Not Available as Promised	260

■ ■ ■ **Fraud Alert: Fraud and scam complaints account for more than 56% of all Money Transfer complaints. Immediate investment in fraud monitoring and transaction verification is critical.**

5.6 Debt Collection

Issue	Complaints
Attempt to Collect Debt Not Owed	1,351
Written Notification Issues	487
Legal Action Threats	380

Many customers dispute debt ownership. Improved debt validation and communication transparency are required to reduce this complaint category.

6. Geographic Risk Analysis

State	Complaints	Risk Level	% of Total
California	13,709	■ Critical	21.93%
Florida	6,488	■ High	10.38%
Texas	4,686	■ High	7.50%
New York	4,442	■ Elevated	7.10%
Georgia	2,921	■ Elevated	4.67%
New Jersey	2,664	■ Moderate	4.26%
Illinois	2,270	■ Moderate	3.63%

- California alone contributes 21.93% of all complaints — 2.1× more than Florida.
- The top 4 states (CA, FL, TX, NY) account for approximately 47% of all complaints.
- Complaints are heavily concentrated in high-population coastal and southern states.
- 51 states covered providing full geographic visibility.

■ **Recommendation:** *Build state-level monitoring dashboards for California, Florida, Texas, and New York — these 4 states represent nearly half of all complaint volume.*

7. Complaint Trend Analysis

7.1 Yearly Complaint Growth (2017–2023)

Year	Complaints	YoY Change
2017	5,394	Baseline
2018	7,872	+45.9%
2019	7,075	-10.1%
2020	8,942	+26.4%
2021	11,149	+24.7%
2022	12,953	+16.2% ← Peak
2023	9,131	-29.5% ← Declining

- Complaint volume grew 140% from 2017 (5,394) to peak in 2022 (12,953).
- 2023 shows a significant 29.5% decline from the 2022 peak — a positive signal.
- The 2020 spike likely reflects pandemic-era financial stress and service disruptions.
- Sustained growth from 2020–2022 may indicate business expansion or emerging service challenges.

7.2 Quarterly Complaint Patterns

Quarter	Complaints	% of Annual Total
Q1	14,820	23.7%
Q2	16,569	26.5% ← Elevated
Q3	16,923	27.1% ← Peak
Q4	14,204	22.7% ← Lowest

■ Q2 and Q3 carry 53.6% of annual complaint volume. Increase support staffing and monitoring capacity during these peak periods to protect SLA performance.

8. Complaint Submission Channel Analysis

Channel	Complaints	Share
Web	45,423	72.66%
Referral	10,766	17.22%
Phone	4,684	7.49%
Postal Mail	1,318	2.11%
Fax	231	0.37%

- 72.66% of complaints submitted digitally via web — confirming digital-first customer behavior.
- Referral (17.22%) is the second-largest channel, indicating strong third-party routing.
- Traditional channels (phone, mail, fax) collectively handle only 9.97% of complaints.
- Investment in digital complaint management and self-service will yield the highest ROI.

9. Power BI Dashboard

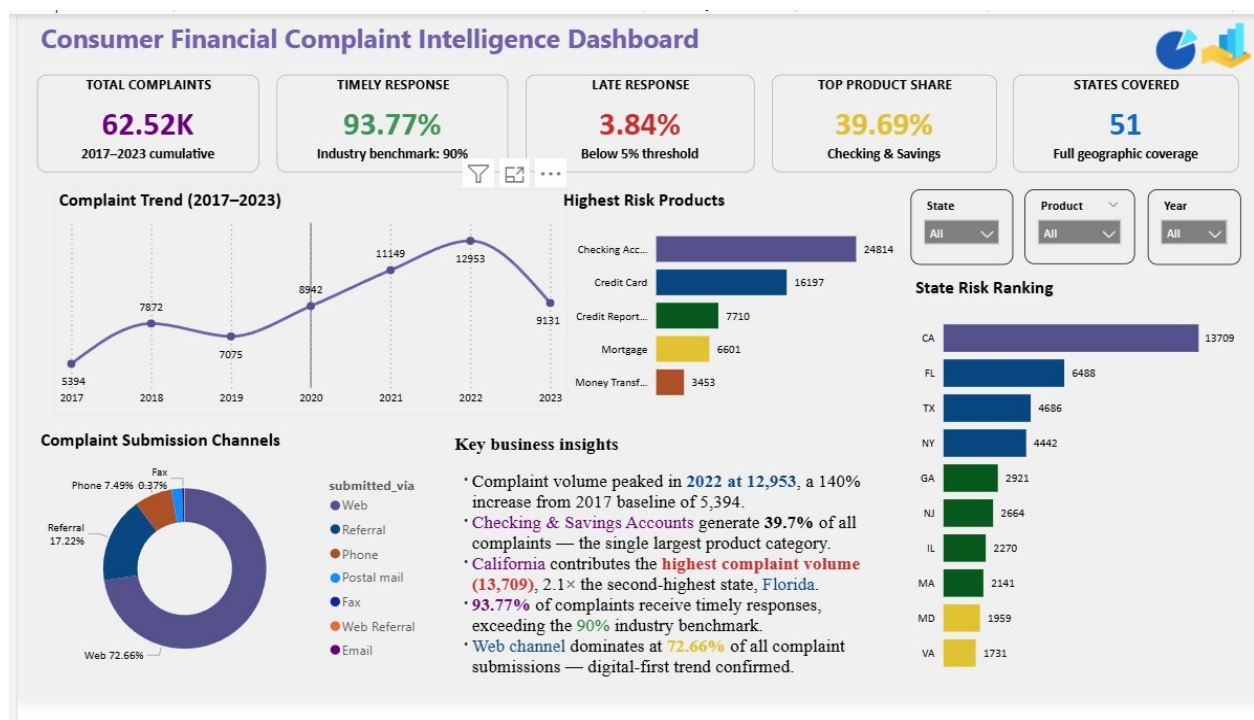


Figure 1: Consumer Financial Complaint Intelligence Dashboard — Power BI (KPI scorecards · Trend line · Product risk ranking · State risk map · Channel donut · Interactive slicers)

Dashboard Features

- KPI Scorecards: Total Complaints, Timely Response %, Late Response %, Top Product Share, States Covered
- Complaint Trend Line (2017–2023): Year-over-year volume with data labels
- Highest Risk Products: Horizontal bar chart ranked by complaint volume
- State Risk Ranking: Bar chart showing top 10 states by complaint count
- Submission Channel Donut: Visual breakdown of all complaint channels
- Interactive Slicers: Filter entire dashboard by State, Product, and Year
- Live Dashboard: <https://app.powerbi.com/groups/me/reports/d8338132-ed69-4cdf-8f4d-465d75ab7989>

10. Strategic Recommendations

1. Account Management Overhaul

Audit and redesign account open/close/manage workflows. 15,109 complaints on account management alone represent 60%+ of all checking complaints. Streamlined digital self-service flows should be the priority.

2. Credit Card Onboarding Redesign

1,867 'Getting a Credit Card' complaints indicate acquisition funnel friction. Simplify the application and onboarding journey to reduce early-stage dissatisfaction.

3. Fraud Detection Investment

Deploy enhanced real-time transaction monitoring for Money Transfer products immediately. Fraud accounts for 56%+ of the category's complaints — this is a regulatory and reputational risk.

4. Credit Reporting Accuracy

4,145 incorrect information complaints demand stronger data governance. Improve dispute resolution SLAs and data integrity checks with credit bureaus.

5. Q2–Q3 Staffing Plan

These two quarters carry 53.6% of annual complaint volume. Build a seasonal staffing model to protect 93.77% timely response rate during peak periods.

6. Regional Monitoring Dashboards

Build dedicated state-level dashboards for California, Florida, Texas, and New York. These 4 states represent ~47% of all complaints and require focused regional oversight.

7. Digital Channel Investment

72.66% of complaints arrive via web. Continue investing in digital complaint management, AI-assisted triage, and self-service resolution capabilities.

11. Skills Demonstrated

Skill	Applied How
SQL	5-file query suite: DDL, DML, window functions (DENSE_RANK, SUM OVER), aggregations, C
Python / Pandas	Data cleaning pipeline — null handling, dtype casting, date parsing, feature engineering
Data Modeling	Flat table schema with engineered time dimensions (month, year, quarter, resolution_days
KPI Design	Defined and computed 9 business KPIs with formulas, benchmarks, and business context
Data Storytelling	Structured findings: Problem → Finding → Business Impact → Recommendation
Power BI	Executive dashboard with KPI cards, trend lines, risk rankings, donut charts, slicers
Business Acumen	Translated complaint data into product risk scores, geographic alerts, staffing strategy
Documentation	Project charter, data dictionary, KPI definitions, SQL insights report, README

12. Conclusion

This project delivers a full-stack analytics solution transforming 62,516 raw consumer complaint records into structured intelligence for financial decision-makers. The analysis reveals that Checking & Savings Accounts and Credit Card Products are the primary complaint drivers, accounting for 65.6% of total volume. Account management friction, transaction disputes, inaccurate credit reporting, and money transfer fraud represent the most actionable customer pain points.

While the 93.77% timely response rate demonstrates strong operational discipline — exceeding the 90% industry benchmark — the sustained complaint growth from 2017 to 2022 (140% increase) signals underlying product and process issues that require strategic investment. The 2023 decline to 9,131 complaints suggests early-stage improvement, but continued monitoring of high-risk products and states remains essential.

The seven strategic recommendations outlined in this report provide a clear roadmap: from immediate fraud monitoring investments to seasonal staffing models and regional dashboards. Executing these recommendations systematically will substantially reduce complaint volumes, improve customer satisfaction, and strengthen regulatory compliance posture.

■ ***This project demonstrates end-to-end data analytics capability: business problem framing, data engineering, SQL analysis, KPI design, executive visualization, and business storytelling.***